API Integration and Schema Adjustment Report

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E-commerce MarketPlace Report

This report explains the steps to integrate APIs, adjust schemas in Sanity CMS, and display data on the frontend.

API Integration Process:

GET /product: Get the list of products.

GET /category: Fetch product categories.

Data Migration:

```
}
Category created successfully {
  _createdAt: '2025-01-21T07:45:12Z',
  _id: 'crockory-24',
  _rev: 'H4lKdx2AthRpksOOBpdEnC',
  _type: 'category',
  _updatedAt: '2025-01-21T07:45:12Z',
  name: 'Crockory',
  slug: 'crockory'
}
Uploading product: {
  _id: 'product-24',
  _type: 'product',
  name: 'Marble Ease',
  slug: { _type: 'slug', current: 'marble-ease' },
  price: 419,
  category: { _type: 'reference', _ref: 'crockory-24' },
  tags: [],
  quantity: 50,
  image: {
    _type: 'image',
    asset: {
      _type: 'reference',
      _ref: 'image-7f6146b6625bf94030f2aa896e9b0c847e5f4919-8000x8000-jpg'
    }
  },
  description: 'A timeless design, with premium materials features as one of our most popular
 with beech legs and lambskin leather upholstery.',
  features: [
    'Premium material',
    'Handmade upholstery',
    'Quality timeless classic'
  ],
  dimensions: {
    _type: 'dimensions',
    width: '75cm',
    height: '110cm',
    depth: '50cm'
  }
}
Imported product: Marble Ease
Data import completed!

D:\next.js\hackathon-template02\sanity-migration>
```

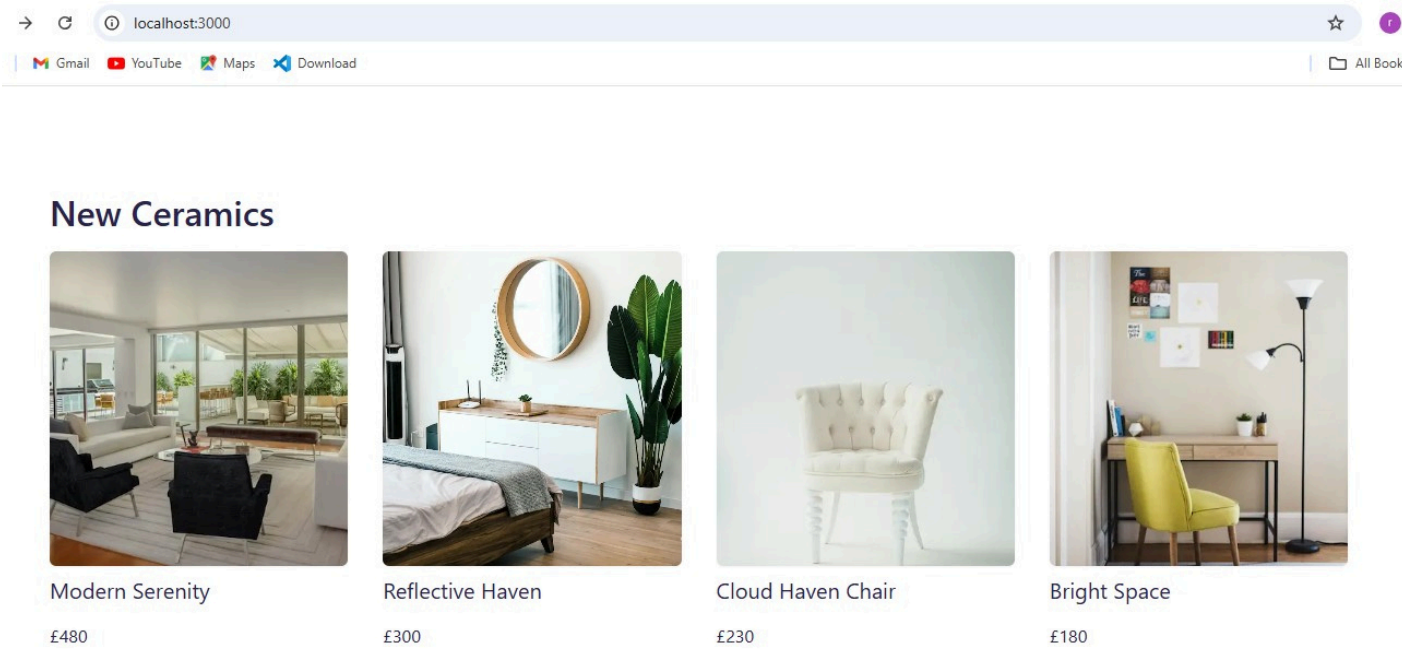
Fetch Data from API:

```
const Ceremics = () => {
  const [products, setProducts] = useState<Product[]>([]);
  const [visibleCount, setVisibleCount] = useState(4); // Controls how many products are visible
  const productsPerPage = 4; // Number of products to load each time

  useEffect(() => {
    const fetchProducts = async () => {
      const query = `*[_type == "product" && category->name in ["Ceramics","Plant Pots"]]{
        _id,
        name,
        price,
        "imageUrl": image.asset->url
      }`;

      try {
        const result: Product[] = await client.fetch(query);
        setProducts(result);
      } catch (error) {
        console.error("Error fetching products:", error);
      }
    };

    fetchProducts();
  }, []);
```

Show Data on Frontend:

```
const Ceremics = () => {

  return (
    <div id="ceramics" className="w-full  md:px-20 p-10 text-[#2A254B]">
      <div className="text-2xl font-semibold sm:text-[32px] my-5 text-[#2A254B]">
        New Ceramics
      </div>
      <div className="text-[#2A254B] grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-8">
        {products.slice(0, visibleCount).map((prod) => (
          <div className="space-y-4" key={prod._id}>
            <Link href={`/product/${prod._id}`}>
              <Image
                src={prod.imageUrl}
                alt={prod.name}
                height={375}
                width={305}
                className="object-cover h-72 rounded-md shadow-sm hover:shadow-xl transition-all mb-2"
              />
              <p className="text-xl">{prod.name}</p>
            </Link>
            <p className="text-base">£{prod.price}</p>
          </div>
        ))}
      </div>
      {/* Button to load more products or show less */}
      <div className="flex justify-center items-center mt-16">
        {products.length > productsPerPage && (
          <Button
            variant={"mybutton"}
            className="w-[170px] h-[56px]"
            onClick={handleViewToggle}
          >
            {visibleCount >= products.length ? "View Less" : "View Collection"}
```

Testing the API:



Schema:

```
import { defineType, defineField } from "sanity"

export const product = defineType({
    name: "product",
    title: "Product",
    type: "document",
    fields: [
        defineField({
            name:"category",
            title:"Category",
            type:"reference",
            to:[{
                type:"category"
            }]
        }
        ),
        defineField({
            name: "name",
            title: "Title",
            validation: (rule) => rule.required(),
            type: "string"
        }),
        defineField({
            name: "slug",
            title: "Slug",
            validation: (rule) => rule.required(),
            type: "slug"
        }),
        defineField({
            name: "image",
            type: "image",
            validation: (rule) => rule.required(),
            title: "Product Image"
        }),
        defineField({
            name: "price",
            type: "number",
            validation: (rule) => rule.required(),
            title: "Price",
```

```
defineField({
    name: "quantity",
    title: "Quantity",
    type: "number",
    validation: (rule) => rule.min(0),
  }),
defineField({
    name: "tags",
    type: "array",
    title: "Tags",
    of:[{
        type: "string"
    }]
}),
defineField({
    name: 'description',
    title: 'Description',
    type: 'text',
    description: 'Detailed description of the product',
  }),
  defineField({
    name: 'features',
    title: 'Features',
    type: 'array',
    of: [{ type: 'string' }],
    description: 'List of key features of the product',
  }),
  defineField({
    name: 'dimensions',
    title: 'Dimensions',
    type: 'object',
    fields: [
      { name: 'height', title: 'Height', type: 'string' },
      { name: 'width', title: 'Width', type: 'string' },
      { name: 'depth', title: 'Depth', type: 'string' },
    ],
    description: 'Dimensions of the product',
```

Products Displayed on Frontend:

localhost:3000/allproducts

ail YouTube Maps Download

The Poplar suede sofa
$980

Tropical Vibe
$550

Sleek Living
$300

Serene Seat
$350

Nordic Elegance
$280

TimberCraft
$320

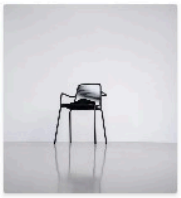
Timeless Elegance
$320

Modern Serenity
$480

Reflective Haven
$300

Sunny Chic
$400

Bold Nest
$260



The Steel Chair
$250